

Genetic Code & Translation

Current 33-page faculty deck primary - official Week 4 objectives integrated - genetic-code logic, translation mechanics, regulatory checkpoints, viral hijacking, and protein targeting

4 official objectives

Mutation logic

A/P/E sites

Kozak vs Shine-Dalgarno

IRE/IRP

Viral IRES

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the same material for retrieval and application without replacing the source deck.

Official Week 4 Objectives

1. Discuss the salient features of the genetic code and distinguish mutation types by their effects on protein sequence and function.
2. Summarize initiation, elongation, and termination of translation and explain major mechanisms regulating eukaryotic protein synthesis.
3. Explain how viruses hijack host translational machinery to favor synthesis of viral proteins.
4. Identify subcellular sites of protein synthesis and explain major post-translational processing and targeting events.

ASSIGNED SCHOLARRX

DNA Transcription & Translation

The code is read in non-overlapping **triplet codons** on mRNA. It is **degenerate** because most amino acids have more than one codon, but it is not ambiguous: each codon specifies one amino acid or stop signal.

Start

AUG establishes the reading frame and encodes methionine; the initiator tRNA is specialized for initiation.

Stop

UAA, UAG, UGA do not bind tRNAs. Release factors recognize them and terminate synthesis.

Direction

Ribosome reads mRNA **5' to 3'**; the polypeptide grows from **N-terminus to C-terminus**.

Code logic: synonymous codons are possible because degeneracy is concentrated at the third codon position, where wobble pairing permits one tRNA to recognize more than one codon.

Mutation	Effect	Protein consequence
Silent / synonymous	Codon changes but amino acid does not	Often no primary-sequence change
Missense	One amino acid replaced by another	Effect ranges from minimal to major, depending on residue and position
Nonsense	Sense codon becomes stop	Premature termination; truncated protein

Frameshift	Insertion/deletion not divisible by 3	Changes downstream reading frame and often introduces an early stop
In-frame insertion/deletion	3n nucleotides added/deleted	Adds/removes amino acids without shifting downstream frame

Exon-splicing connection: deletion of an exon does not automatically mean a frameshift. Whether the downstream frame is preserved depends on the exon boundaries and total number of nucleotides removed.

Slides 11-17

tRNA is the adaptor; aminoacyl-tRNA synthetase is the true translator

LO 2

Amino acid + tRNA

->

aminoacyl-tRNA synthetase + ATP

->

charged aminoacyl-tRNA

->

ribosome

Aminoacyl-tRNA synthetases attach the correct amino acid to the correct cognate tRNA. This charging reaction uses ATP and provides much of translation's fidelity.

Wobble: relaxed base pairing between the 3rd nucleotide of the codon and the 1st nucleotide of the anticodon allows the cell to translate 61 sense codons with fewer distinct tRNAs.

Slides 18-21

Initiation establishes the correct AUG in the P site

LO 2

Prokaryotes	Eukaryotes
30S subunit uses 16S rRNA to base-pair with the Shine-Dalgarno ribosome-binding site.	40S complex is recruited near the 5' cap and scans to an AUG in a favorable Kozak sequence.
Initiator amino acid is fMet .	Initiator amino acid is Met .
Initiation factors assemble 30S + 50S into 70S.	Multiple eIFs assemble 40S + 60S into 80S.

Clinical anchor from the deck: Kozak-sequence mutations can reduce translational efficiency; the lecture uses beta-globin/Kozak changes as a beta-thalassemia example.

Slides 22-23

Elongation cycles through A, P, and E sites; release factors terminate at stop codons

LO
2

AA-tRNA enters A site

->

peptide bond forms

->

ribosome translocates

->

deacylated tRNA exits E site

Peptidyl transferase activity resides in ribosomal rRNA - the ribosome is a ribozyme. At a stop codon, release factors enter instead of a tRNA and promote hydrolysis of the completed polypeptide.

Slides 24-27

Eukaryotic translation is regulated at initiation and at specific mRNAs

LO 2

eIF2

Phosphorylation of eIF2 α inhibits GDP-to-GTP recycling through eIF2B and suppresses global initiation. The deck uses heme-regulated control of globin synthesis as a model.

eIF4E

4E-BP can sequester cap-binding eIF4E. Phosphorylation of 4E-BP releases eIF4E and promotes cap-dependent translation.

Iron-response elements (IREs)

At **low iron**, IRP binds IREs: 5' UTR binding inhibits translation of ferritin/ALA synthase, while 3' UTR binding stabilizes transferrin-receptor mRNA. At **high iron**, IRP binds iron and releases the RNA: storage proteins rise while transferrin-receptor mRNA becomes less stable.

Low Fe: store less, import more. **High Fe:** store more, import less.

Slides 28-29

Viruses redirect host ribosomes toward viral RNAs

LO 3

IRES strategy: an internal ribosome entry site can recruit translation machinery without normal 5' cap-dependent initiation. This gives some viral RNAs a way to keep being translated while host cap-dependent synthesis is suppressed.

Core concept: viruses do not bring a separate human-style translation system. They exploit host ribosomes and initiation machinery, then alter the rules to favor viral transcripts.

Slides 30-32

New polypeptides must fold, modify, and reach the correct compartment

LO 4

Free cytosolic ribosomes

Make proteins that remain in cytosol or are imported post-translationally into nucleus, mitochondria, peroxisomes, and related destinations.

Rough ER-bound ribosomes

Make secreted proteins, lysosomal proteins, and many membrane proteins through co-translational ER import.

Chaperones/chaperonins help nascent proteins acquire the correct conformation. Post-translational processing can include cleavage, covalent modification, glycosylation, phosphorylation, acetylation, and trafficking driven by signal sequences.

Signal-sequence handoff: a hydrophobic N-terminal signal can direct a translating ribosome to the ER. This sets up Wednesday Lecture 9, where the trafficking machinery is developed in detail.

Exam traps

Protect code, initiation, and regulation relationships

All LOs

Trap 1: Wobble is codon-anticodon flexibility, not an error in the genetic code.

Trap 2: Shine-Dalgarno = bacteria; Kozak = eukaryotes.

Trap 3: A frameshift requires an insertion/deletion not divisible by 3.

Trap 4: eIF2 phosphorylation suppresses initiation; 4E-BP binding suppresses eIF4E.

Trap 5: IRES permits cap-independent translation; it does not create a new ribosome.

30-SECOND FINISH

AUG starts; UAA/UAG/UGA stop. Synthetases charge tRNAs. Shine-Dalgarno finds AUG in bacteria; Kozak helps eukaryotes. A/P/E sites drive elongation. eIF2-P turns global initiation down. IRE/IRP ties translation to iron. IRES lets viral RNA bypass cap dependence.